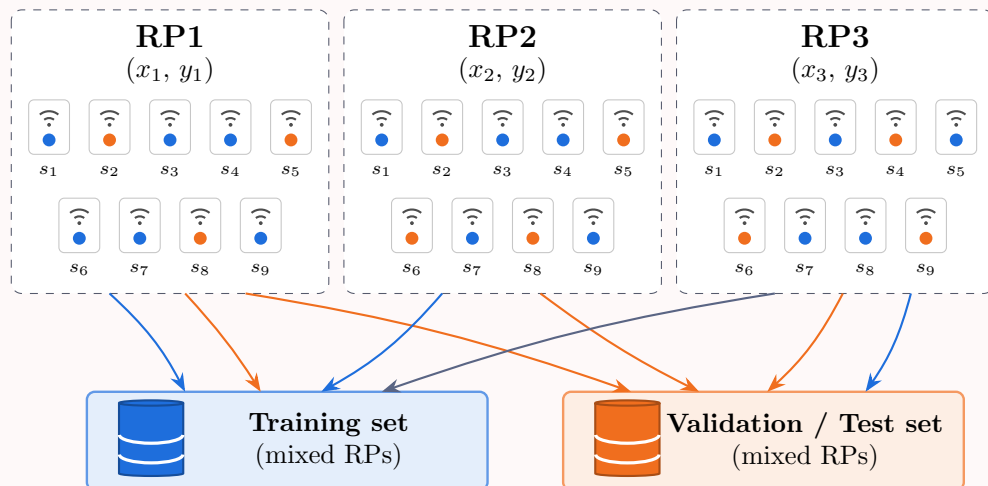


A

Ordinary Random Sample Split



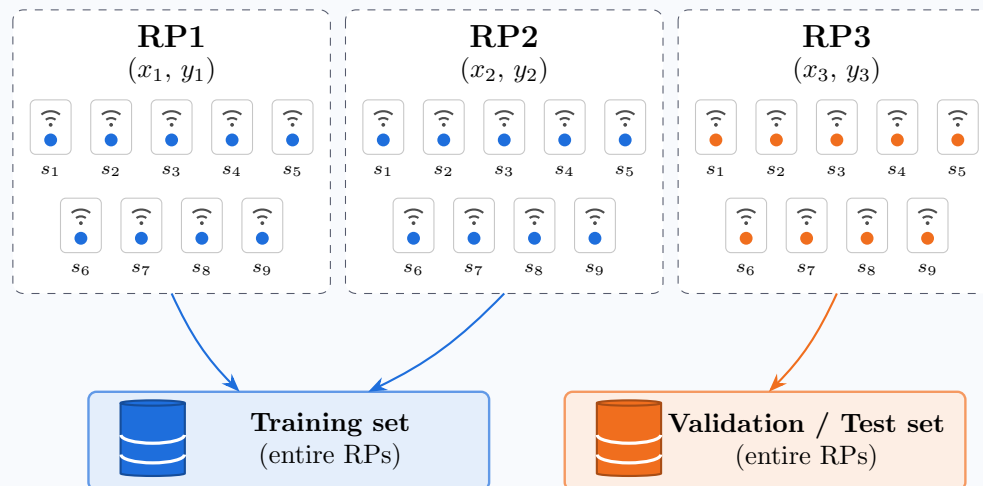
Same physical RP appears in both TRAIN and VALIDATION

- Spatial leakage / optimistic evaluation
- $(x_i, y_i) = (x_j, y_j)$, but samples are split across folds

Easier evaluation, inflated performance

B

Proposed RP-Grouped Validation



All scans from one physical RP stay in the same fold

- No spatial leakage
- $g_i = g_j$ iff $(x_i, y_i) = (x_j, y_j)$

GroupKFold-style split (example)

Fold 1
Train: {RP1, RP2}
Test: {RP3}

Fold 2
Train: {RP1, RP3}
Test: {RP2}

Stricter evaluation, better spatial generalization

Aspect	Ordinary random split	RP-grouped split
Unit of splitting	individual scans	entire physical RPs
Leakage risk	high	none across the same RP
Difficulty	easier	stricter
Generalization meaning	repeated measurements at seen RPs	unseen physical locations / stronger spatial generalization



Core idea: grouping repeated scans by physical reference point prevents leakage and produces a more realistic estimate of localization performance.